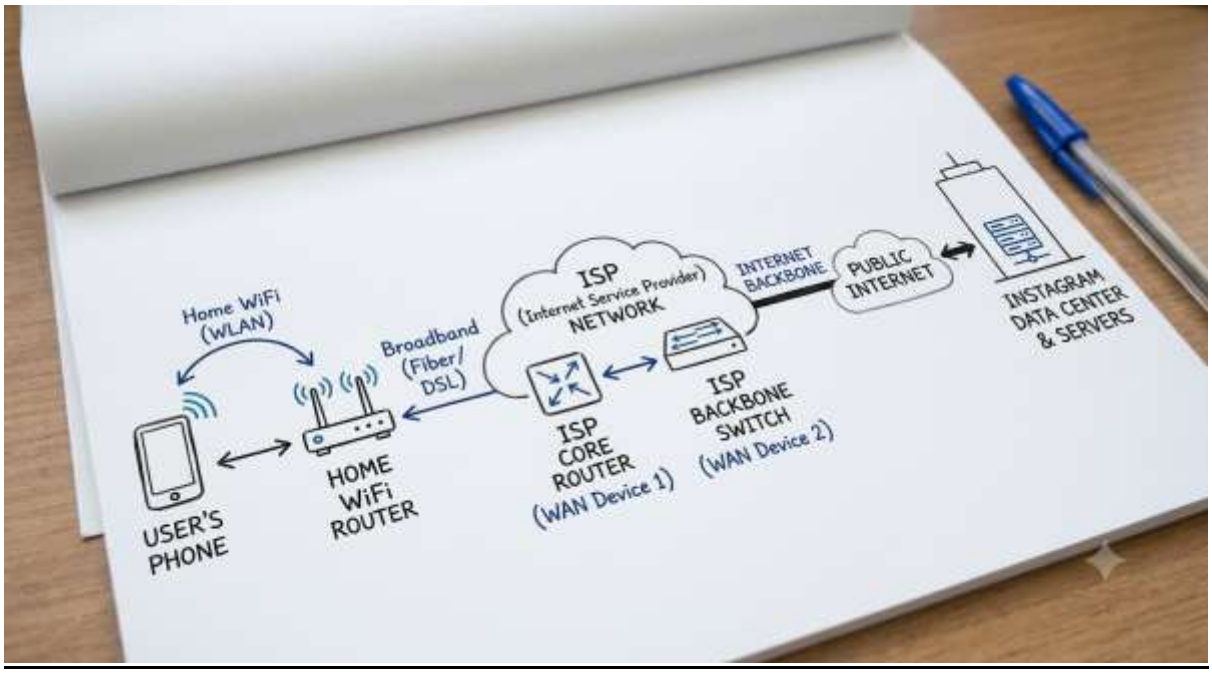
1.

Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

ANS:-



2.

Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

ANS:-

WAN Technology	Speed / Performance	Cost	Real-World Use Case
MPLS (Multiprotocol Label Switching)	High, highly reliable, low latency, and predictable performance.	High / Expensive	Connecting core data centers and central servers for critical transactional systems like IRCTC ticketing.
Leased Line (Dedicated Internet Access)	Dedicated symmetrical speeds (equal upload and download), zero contention.	Very High	Corporate regional offices requiring guaranteed uptime and heavy data exchange, such as a Zomato corporate office.
Broadband Internet (Fiber / Cable)	High download, but variable/asymmetrical upload speeds; shared medium.	Low / Affordable	Residential usage, streaming, and remote work at your home.

3.

Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely

from a public WiFi at a coffee shop. Hint: Focus on how the VPN creates a private tunnel over the public WAN.

ANS:-

When you connect to public Wi-Fi at a coffee shop, your data travels across public Wide Area Network (WAN) infrastructure where malicious actors can easily intercept unencrypted traffic.

A Virtual Private Network (VPN) solves this by creating an encrypted "tunnel" over the public WAN:

1. Encapsulation & Encryption: When you open Spotify or YouTube, the VPN client on your phone or laptop scrambles your data packets into encrypted code before they leave your device.
2. Tunneling Across Public WAN: This encrypted data is transmitted across the coffee shop's public Wi-Fi and the ISP's public WAN. To any snooping hackers or routers along the path, the data looks completely unreadable.
3. Decryption at VPN Server: The traffic securely reaches the VPN provider's remote server, where it is decrypted and forwarded to the destination servers (like Spotify or YouTube). This ensures your browsing activity remains completely private and secure from local network snooping.

4.

Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

ANS:-

For an e-commerce giant like Flipkart connecting mission-critical warehouses across major cities, the best WAN technology is **MPLS (Multiprotocol Label Switching)** (or modern **SD-WAN** built over high-speed enterprise connections).

Justification (Two Reasons):

1. **Guaranteed Reliability and Low Latency:** Warehouse operations rely on real-time inventory updates, barcode scanning, and fast dispatch logs. MPLS provides dedicated routing paths with strict Quality of Service (QoS) guarantees, preventing packet drops and delays even during high-traffic festival sales (like Big Billion Days).
2. **High Security for Enterprise Data:** Since warehouses handle sensitive supply chain data, internal logistics, and customer order details, MPLS operates over private routing infrastructures rather than the open public internet, significantly reducing exposure to cyber threats and interception.